

Chapter 15

Cleaning Using High-Speed Impinging Jet

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15.1 INTRODUCTION

Surface cleaning using a high-speed air jet can be applied for the removal of solid particulate contaminants adhering to a solid surface. The procedure is simple. An air jet is applied to the surface and the air blows off the particle contaminant. The physical mechanism causing the removal is resuspension of particles from a solid surface. The resuspension phenomenon has been studied by means of tube flow. Many experimental and theoretical studies have been reported and are well summarized by Ziskind et al.¹ However, the model of the resuspension phenomenon has not yet been established. Thus, the removal efficiency of the high-speed air jet method cannot be estimated theoretically.

Therefore, the present chapter focuses on empirical knowledge. The author has been carrying out several studies with particular attention to the effect of operating conditions on the removal efficiency.² The results and discussion will be summarized in this chapter. In addition, new removal methods based on the air jet removal method will be introduced.

The data given in this chapter were obtained from experiments using mono-dispersed standard latex particles (styrene/divinylbenzene) as test particles and borosilicate glass as a standard surface. The test particles were deposited by sedimentation in air. After deposition, the test piece was dried for more than 100 h in a desiccator. For all experiments, the test piece was transferred to a controlled removal environment, and the removal experiment was conducted after leaving the test piece in the environment for 2 min or longer. The experimental procedure will be explained in [Section 15.2.1](#).

15.2 FUNDAMENTALS OF AIR JET REMOVAL

15.2.1 Apparatus and Parameters^{3,4}

[Figure 15.1](#) shows the apparatus for removal using a high-speed air jet. The air fed by a compressor usually contains water mist, which is separated before it is passed to a pressure regulator and a jet nozzle. The air jet nozzles used in the following experiments are shown in [Figure 15.2](#). The flow through the conduit in the nozzle is suddenly reduced near the nozzle outlet and the pressure drop caused by the flow contraction is the highest in the air jet system. The pressure drop at the nozzle tip, $\Delta P_n (=P_n - P_a)$, where P_n is the air pressure in the nozzle and P_a is the ambient air pressure), is almost the same as that set by use of the pressure regulator. The pressure drop ΔP_n is one of the main operating conditions.

[Figure 15.3](#) shows the configuration of the nozzle and the surface on which the particles are deposited. The geometric parameters for removal are the impinging angle θ between the nozzle centerline and the surface, and the distance d between the nozzle tip and the point O on the surface where the nozzle

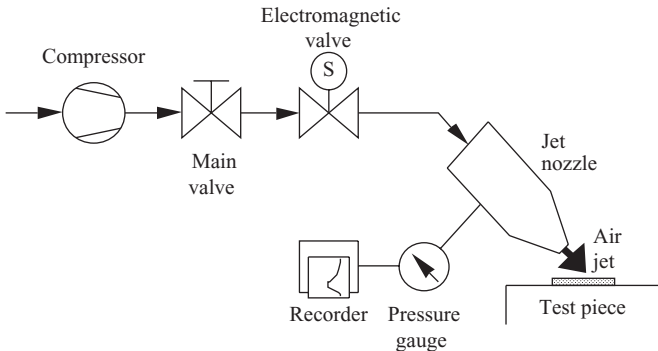


FIGURE 15.1 Apparatus for particle removal using the impinging air jet.

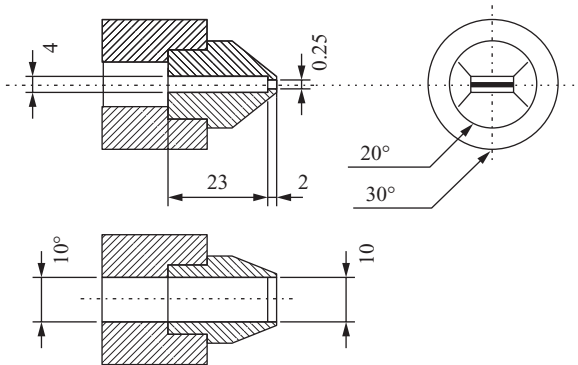


FIGURE 15.2 Schematic view of the air jet nozzle (dimensions are in millimeters).

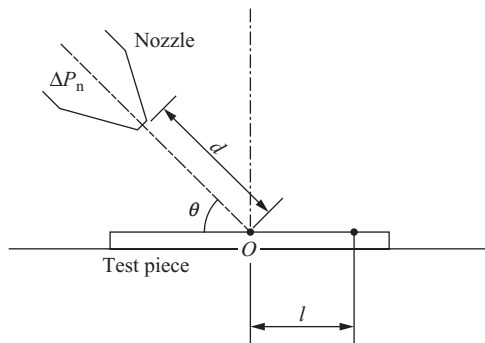


FIGURE 15.3 Geometric parameters for arrangement of the nozzle and the surface.

centerline crosses the surface. In the following sections, the effect of these parameters on the removal efficiency will be described.

Before describing the removal efficiency, the characteristics of the air jet are presented. Figure 15.4 shows the change in the pressure drop ΔP_n with time and the corresponding dynamic pressure of the air jet P_d at various distances d . The electromagnetic valve has a time lag of about 0.4 s and the pressure change is also delayed from the “on” and “off” signals of the timer. The pressures ΔP_n and P_d show peak values just after valve opening. The peak values are, however, less than 3% of the steady-state pressure value. Therefore, the pressure change is well approximated by a step function and the air jet is well developed for the entire region of $d = 5\text{--}20$ mm within about 0.2 s.

Figure 15.5 shows the change in dynamic pressure P_d along the centerline of the air jet for a pressure drop $\Delta P_n = 10^5$ Pa and 3×10^5 Pa gauge pressure. The dynamic pressure P_d for $d > 7$ mm is almost inversely proportional to the distance d , which means that the flow velocity is proportional to $d^{-1/2}$. This confirms that the jet is fully developed.

When the air jet is assumed to be a two-dimensional free jet, the air velocity u_0 and the static pressure P_0 are constant in the potential core that is established

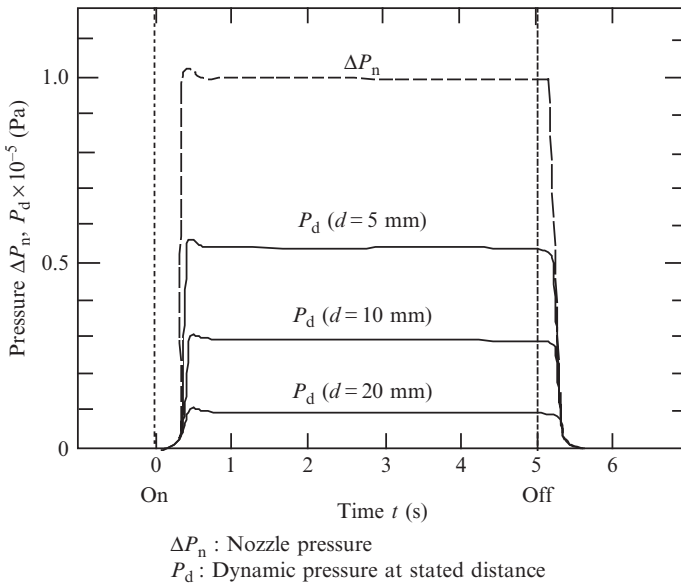


FIGURE 15.4 Change in nozzle pressure drop with time and corresponding dynamic pressure values at various distances.

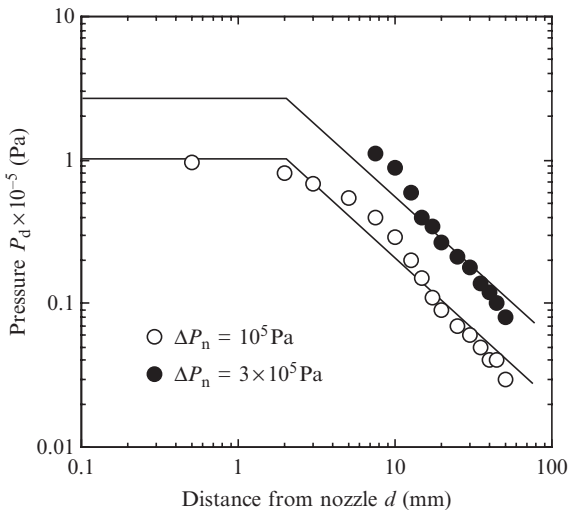


FIGURE 15.5 Dynamic pressure as a function of the distance d . The solid lines are calculated values from Equations (15.2) to (15.7).

within $d < 5b-8b$ (where b is the nozzle gap) from the nozzle tip. It is assumed that the flow in the nozzle is a polytropic process with friction loss at the wall. Thus, the air velocity at the nozzle tip u_0 is calculated by the following equation:

$$u_0 = \sqrt{\frac{2\kappa}{\kappa-1}RT_0 \left\{ 1 - \left(\frac{P_n}{P_0} \right)^{(n-1)/n} \right\}} \quad (15.1)$$

where R is a constant ($R = 287 \text{ kgf-m}^2/\text{kgf-K-s}^2$ for air), κ is the ratio of specific heats, and n is the polytropic index. The polytropic index n is related to the velocity coefficient ϕ of the nozzle by the following equation:

$$\phi = \sqrt{\left\{ 1 - \left(\frac{P_n}{P_0} \right)^{(n-1)/n} \right\} \left\{ 1 - \left(\frac{P_n}{P_0} \right)^{(\kappa-1)/\kappa} \right\}^{-1}} \quad (15.2)$$

It is known that the coefficient ϕ for a nozzle having a well-finished wall is in the range 0.95-0.975.

When the nozzle pressure P_n reaches a critical pressure P_{nc} , calculated by the following equation, the air velocity attains the maximum velocity:

$$\frac{P_{nc}}{P_a} = \left(\frac{2}{n+1} \right)^{-n/(n-1)} \quad (15.3)$$

When the nozzle pressure exceeds the critical pressure P_{nc} , the air pressure, P'_0 , at the nozzle tip is larger than the ambient air pressure P_a :

$$P'_0 = P_n \left(\frac{2}{n+1} \right)^{n/(n-1)} \quad (15.4)$$

The high-pressure air expands in the potential core region. Because momentum is conserved during the expansion, the air velocity u_0 after the expansion can be expressed as follows:

$$u_0 = \sqrt{\frac{P_0}{P_a} u_0'^2 + \frac{P'_0 - P_a}{\rho_a}} \quad (15.5)$$

where u'_0 is the air velocity at the critical pressure ($P_n = P_{nc}$) and ρ_a is the density of air at ambient air pressure.

As shown in [Figure 15.5](#), the air velocity in the fully developed region is proportional to $d^{-1/2}$. By assuming that the imaginary origin of the jet is at the nozzle tip, the air velocity at distance d can be expressed by

$$u(d) = \sqrt{\frac{K_u}{d}} u_0 \quad (15.6)$$

where K_u is a proportionality constant.

The nozzle (Figure 15.2) has a nozzle gap, $b = 0.25$ mm. Therefore, it is estimated that the length of the potential core is $d = 2.0$ mm ($=8b$). At the end of the potential core, the air velocity $u(d)$ is equal to u_0 . Thus, the constant K_u is determined to be 2.0×10^{-3} . By substituting K_u into Equation (15.6), the air velocity under various operating conditions can be estimated. The lines in Figure 15.5 show the estimated dynamic pressure $P_d (= \rho_a u(d)^2/2)$. For the calculation, the velocity coefficient ϕ was assumed to be 0.975. The lines fit the data well. It should be noted that the constant ϕ represents the nozzle characteristics given here. This means that different values should be adopted for different shapes and sizes of the nozzle.

15.2.2 Definition of Removal Efficiency

It is well known that the resuspension flux changes with time.^{5,6} The removal efficiency obtained by the impinging air jet also changes with time, as shown in Figure 15.6. In this figure, two removal efficiencies are defined, the instantaneous removal efficiency $\eta(t)$ and the integrated removal efficiency $\Sigma\eta(t)$:

$$\eta(t) = (\sigma_p(t + \Delta t) - \sigma_p(t)) / \sigma_{p0} \quad (15.7)$$

$$\Sigma\eta(t) = (\sigma_p(t) - \sigma_{p0}) / \sigma_{p0} \quad (15.8)$$

where $\sigma_p(t)$ is the number density of deposited particles on a surface at time t and σ_{p0} is the initial number density.

In some cases, the final removal efficiency was reached within the time resolution of 150 ms after the jet starts. In other cases, after a quick initial change (first step removal), the integrated removal efficiency further increased with jet exposure time and finally reached saturation after typical times of 1 s. The removal efficiency during steps 1 and 2 depends on the air jet condition. The mechanism of removal in each step is under investigation. Therefore, in this chapter the focus will be on the saturated removal efficiency. The saturated removal efficiency is simply defined as the removal efficiency η .

15.2.3 Effect of Operating Conditions on the Removal Efficiency η

15.2.3.1 Pressure Drop ΔP_n and Distance d ⁴

Figure 15.7 shows the removal efficiency η for various particles as a function of the pressure drop ΔP_n at the nozzle tip. The removal efficiencies increase with the pressure. When the distance d is longer, higher pressure is required to achieve the same removal efficiency. In order to evaluate the difference caused by the distance, the dynamic pressure at distance d was used as a representative value of the energy of the air jet. Figure 15.8 shows the removal efficiency for a particle with $D_p = 5.1$ μm as a function of the dynamic pressure P_d . Here, the air

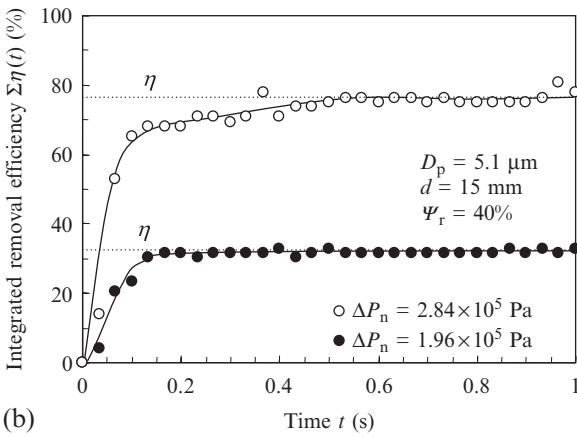
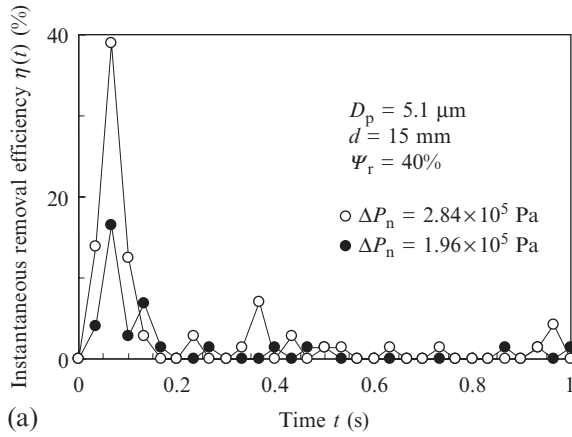


FIGURE 15.6 Change in the (a) instantaneous and (b) integrated removal efficiencies with time.

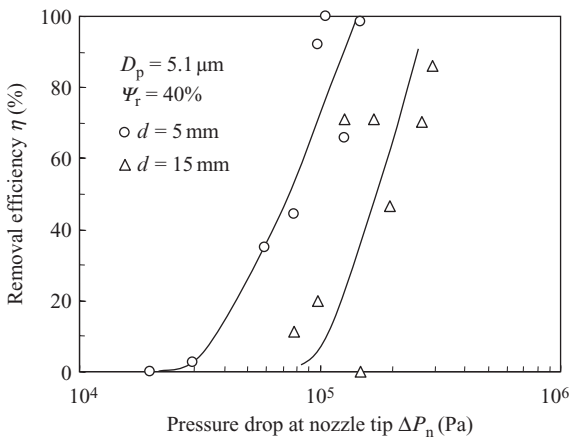


FIGURE 15.7 Removal efficiency as a function of the pressure drop ΔP_n .

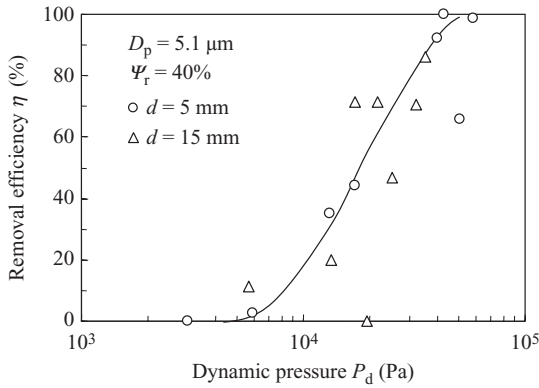


FIGURE 15.8 Removal efficiency replotted as a function of the dynamic pressure P_d .

velocity at distance d was estimated from Equation (15.6). The removal efficiency is well correlated with the dynamic pressure and is independent of the distance and pressure drop. Using this figure, the removal efficiency at other distances can be estimated.

Figure 15.9 shows the performance curve for particles of different sizes. It is obvious that smaller particles require higher dynamic pressure. The curve for a particle with $D_p = 6.4 \mu\text{m}$ shows a slower increase than for other particles, which may be because the test particles have a relatively wider size distribution than the other particles. This implies that the performance curve depends on both the nozzle and the adhesive force distribution.

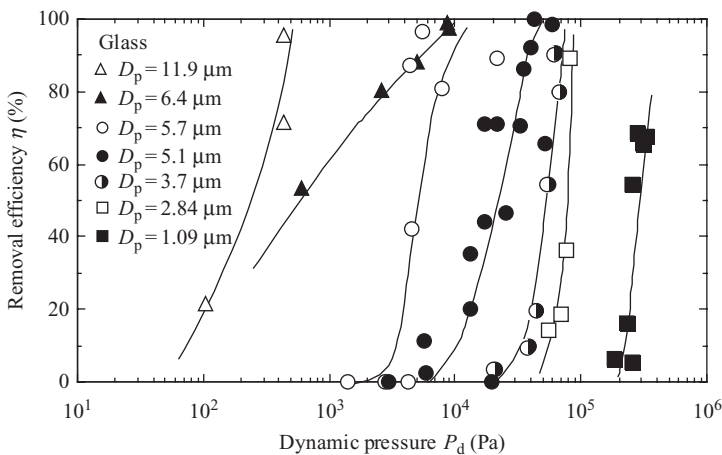


FIGURE 15.9 Removal efficiency as a function of the dynamic pressure P_d for different particle sizes.

15.2.3.2 Impinging Angle $\theta^{3,7}$

Figure 15.10 shows the removal efficiency for an impinging angle (θ) of 45° as a function of the distance l from the impinging point O (see also Figure 15.3). The other parameters are the nozzle pressure drop $\Delta P_n = 10^5$ Pa, the distance $d = 10$ mm, and the duration of the jet $t = 10$ s. The data denoted by runs 1-3 showed wide scatter. However, near the impinging point O ($l = 0$), the removal efficiency η is high and the repeatability of the experiment is good. The efficiency η upstream of the point O ($-10 \text{ mm} < l < 0$) decreases sharply, and the particle detachment by the jet is unsure for the region $l < -15$ mm. In the downstream location, the efficiency η is higher than that found in the upstream location. However, the region showing high removal efficiency is restricted only to a small part around the jet impinging point ($l < 3$ mm). Particle removal by the jet again shows wide variability for the region $l > 3$ mm. For larger impinging angles, the tendency of the data was similar to that obtained here for $\theta = 45^\circ$.

Figure 15.11 shows the removal efficiency η obtained by setting the impinging angle at 30° and 15° . The high-efficiency region around the jet impinging point is wider than that for $\theta = 45^\circ$. Furthermore, the data for $\theta = 30^\circ$ show a relatively low-efficiency region between $5 \text{ mm} < l < 10$ mm. The high-efficiency region of $l < 5$ mm can be attributed to the direct jet impingement as in the case of $\theta = 45^\circ$. The reason for the high removal efficiency found in the region $l > 10$ mm will be discussed later. As for the high-efficiency region around the jet impinging point, the region for $\theta = 15^\circ$ covers up to about $l = 10$ mm and is wider than that for $\theta = 30^\circ$.

Next, we will discuss the jet characteristics in order to identify the reason for data variability and to define the rule for determining the high-efficiency region. Figure 15.12 schematically shows the jet impinging on a flat wall, where the x and y coordinates are set as shown in the figure and the jet origin is represented

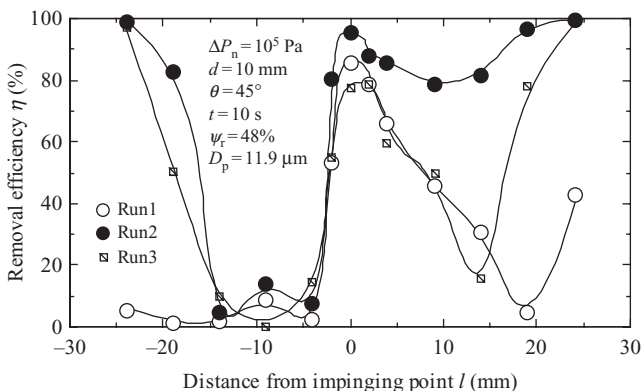


FIGURE 15.10 Removal efficiency for an impinging angle of 45° as a function of the distance from the jet impinging point.



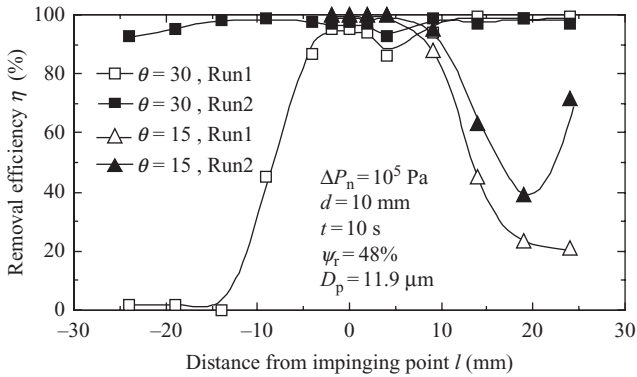


FIGURE 15.11 Effect of impinging angles of 30° and 15° on the removal efficiency.

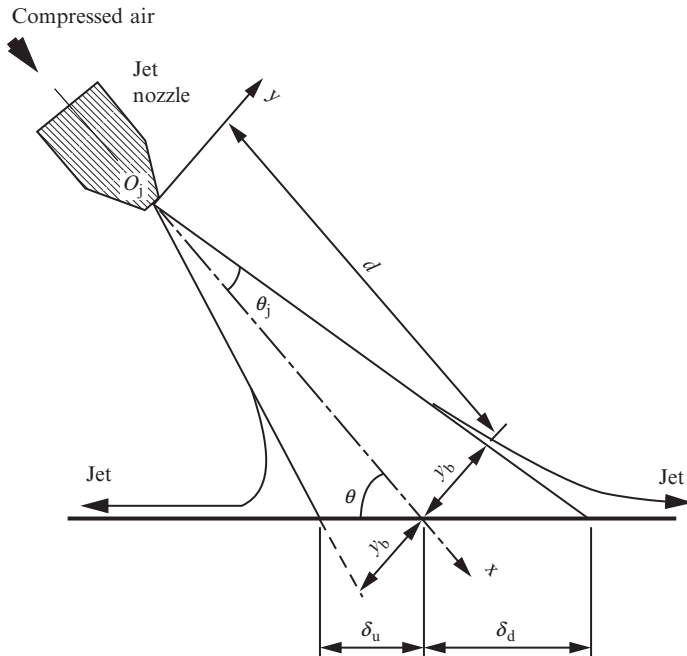


FIGURE 15.12 Schematic diagram of the impinging air jet on a flat wall.

by O_j . As discussed in Section 15.2.2, the impinging point $d = 10$ mm is in the fully developed region of the jet. The effective jet width y is obtained by Tollmien's equation⁸:

$$\phi_j = \frac{y}{K_j x} = 2.4 \quad (15.9)$$

The above equation gives the outer boundary of the two-dimensional jet where the x -component of the flow velocity becomes zero. The empirical constant K_j has a value between 0.09 and 0.12.⁸ Here, we will assume $K_j = 0.1$ as the average. Then the effective jet width y is represented by

$$y = 0.24x \quad (15.10)$$

If we assume that the tip of the nozzle along the centerline is the origin of the jet, the impinging point is given by $x = d$ and the effective jet width y_b is given by

$$y_b = 0.24d \quad (15.11)$$

In Figures 15.10 and 15.11, $d = 10$ mm and therefore y_b is 2.4 mm. The angle of spreading of the jet θ_j is determined to be 13.5° since $\tan \theta_j$ is 0.24 ($= 2.4$ mm/10 mm). The jet expanding width on the flat surface is calculated by the following equations for the upstream and the downstream locations respectively (see also Figure 15.12):

$$\delta_u = d \sin \theta \left\{ \tan \left(\frac{\pi}{2} - \theta \right) - \tan \left(\frac{\pi}{2} - \theta - \theta_j \right) \right\} \quad (15.12)$$

$$\delta_d = d \sin \theta \left\{ \tan \left(\frac{\pi}{2} - \theta + \theta_j \right) - \tan \left(\frac{\pi}{2} - \theta \right) \right\} \quad (15.13)$$

The jet expanding widths calculated by Equations (15.12) and (15.13) are given below:

$$\theta = 45^\circ, \delta_u = 2.73 \text{ mm}, \delta_d = 4.47 \text{ mm}$$

$$\theta = 30^\circ, \delta_u = 3.39 \text{ mm}, \delta_d = 8.22 \text{ mm}$$

$$\theta = 15^\circ, \delta_u = 4.89 \text{ mm}, \delta_d = 89.2 \text{ mm}.$$

The high-efficiency region in Figures 15.10 and 15.11 satisfies the relation $\delta_u < \delta < \delta_d$. Outside this region, the removal efficiency decreases or becomes unstable. The instability may be caused by the turning over and/or separation of the jet from the flat surface. The high efficiency for $\theta = 30^\circ$ and $l > 10$ mm is exceptional, because the flow does not separate from the surface in this case. In this sense, it can be stated that $\theta = 30^\circ$ is the optimum jet angle for detaching small particles from a flat surface without flow separation.

Figure 15.13 shows the removal efficiency around the impinging point as a function of the dynamic pressure P_d for two different particle sizes. When the impinging angle θ is over 30° , the angle does not affect the efficiency for both particle sizes. However, a lower impinging angle θ requires higher dynamic pressure to obtain the same efficiency. In the removal process, not only the removable area discussed above but also the high efficiency is important. Therefore, the removal index N_r is defined as follows:

$$N_r = \eta W (\delta_u + \delta_d) \quad (15.14)$$

where W is the width of the nozzle.

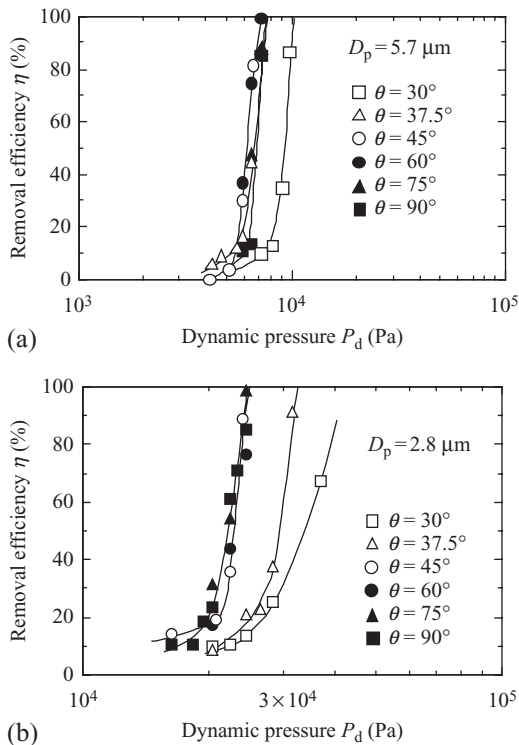
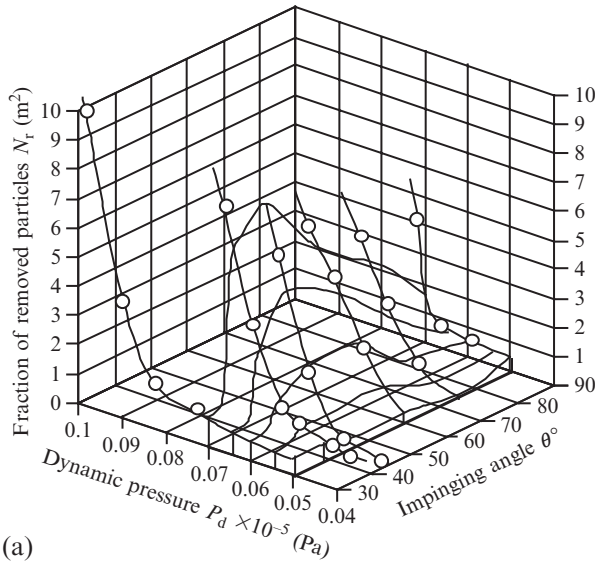


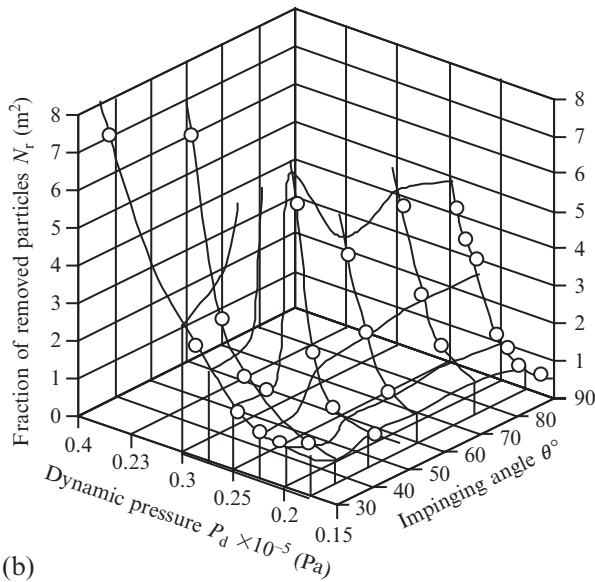
FIGURE 15.13 Effect of the impinging angle on the removal efficiency for (a) $D_p = 5.7 \mu\text{m}$ and (b) $D_p = 2.8 \mu\text{m}$.

The product of the initial number density of deposited particles and the removal index defines the number of particles removed around the jet impinging point. Figure 15.14 shows the removal index plotted against the dynamic pressure and the impinging angle. The lines are equipotential lines. Peaks are observed around the impinging angle 45° in both plots, which suggests that 45° is the optimum angle for obtaining a wide area with high removal efficiency. When the angle is greater than 45° , the removal area ($=W(\delta_u + \delta_d)$) decreases with increasing angle, although the efficiency is not affected by the angle. When the angle is smaller than 45° , the efficiency decreases with decreasing angle, although the removal area increases. Because of both these effects, 45° is the optimum impinging angle.

In Figure 15.13, it was found that an impinging angle greater than 45° did not affect the removal efficiency. However, Ziskind et al.⁹ reported that the removal efficiency decreases for angles greater than 45° and that it shows a peak at an angle of 30° . Thus, the effect of the angle seems to depend on the nozzle shape.



(a)



(b)

FIGURE 15.14 Removal index N_r plotted against the dynamic pressure and the impinging angle: (a) $D_p = 5.7 \mu\text{m}$, $d = 10 \text{ mm}$ and (b) $D_p = 2.84 \mu\text{m}$, $d = 10 \text{ mm}$.

15.2.4 Condition of the Environment^{10,11}

Figure 15.15 shows the result of a removal experiment conducted by varying the humidity ψ_r in the removal environment. The surfaces used in these experiments are borosilicate glass, polybutylene terephthalate, and double nickel-coated iron. For all surfaces, the removal efficiency is low at a low humidity, and rapidly increases as the humidity is increased. When the humidity ψ_d during removal is 67% for a glass plate, the maximum efficiency is achieved, and it decreases for higher humidity. From the figure, it is clear that the removal efficiency can be increased by adjusting the humidity ψ_r in the removal environment. Here, the humidity that achieves the maximum removal efficiency is defined as the optimum humidity ψ_{opt} .

Since the conditions of jet impingement are unchanged, the particle removing force provided by the air jet is considered to be constant. Therefore, it is assumed that the removal efficiency depends on the humidity because water molecules on a particle and the wall surface may have an effect on the adhesion force between the particle and the solid surface. When the humidity is low and the effect of the water molecules is absent, the van der Waals force is the dominant adhesion force. The force can be approximated by a value in vacuum, and represents an adhesion force higher than that in the presence of water molecules. It is well known that the liquid bridging force increases with the humidity, when a liquid bridge is formed between a particle and a surface at high humidity. Thus, it is considered that the removal efficiency is reduced mainly with the increase in the van der Waals force at low humidity, whereas it is reduced with the increased liquid bridge force at high humidity. The maximum removal efficiency is attained at a humidity at which the liquid bridge is formed.

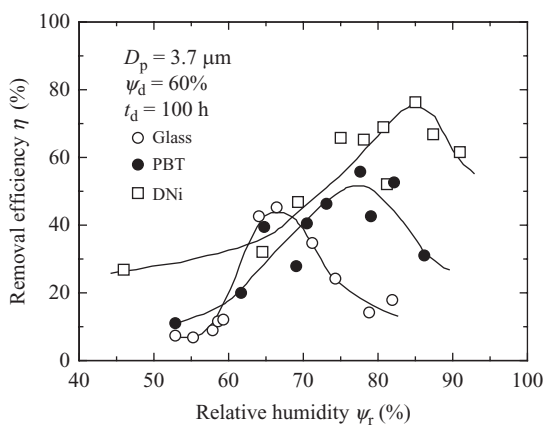


FIGURE 15.15 Removal efficiency from different substrates as a function of the relative humidity in the removal environment.

As shown in Figure 15.15, the optimum humidity ψ_{opt} and the maximum efficiency depend on the surface material. The optimum humidity ψ_{opt} for various surface materials listed in Table 15.1 is shown in Figure 15.16. Some materials used in the experiments have micrometer-sized asperities. The average height and peak count in unit length of the asperities are listed in the table. If the asperity is assumed to be a part of a sphere, the radius of the sphere can be calculated by means of the average height and the peak count. The radius is defined as the surface roughness r_w . The optimum humidity ψ_{opt} is well correlated with the surface roughness r_w .

On the other hand, as shown in Figure 15.17, the maximum removal efficiency has a strong correlation with the removal efficiency at $\psi_r = 55\%$ at which the liquid bridge does not exist. This implies that the adhesion force at the optimum humidity level is dominated by the van der Waals force. Thus, the difference in the maximum removal efficiencies between surface materials must be attributed to the surface properties such as Hamaker’s constant, surface roughness, and stiffness. Phares et al.¹² have studied the effect of surface material

TABLE 15.1 Average Height of Asperities and Peak Count for Different Surface Materials Tested				
	Average Height of Asperities		Peak Count	
	Average R_a (μm)	Standard Deviation σ_r (μm)	Average R_a (μm)	Standard Deviation σ_r (μm)
<i>Metals</i>				
Double nickel-coated iron	0.513	0.19	0.44	0.12
Nickel-coated iron	0.46	0.23	0.23	0.083
Zinc-coated iron	0.062	0.015	1.13	0.35
<i>Plastics</i>				
Polycarbonate	0	0	0	0
Poly(butylene terephthalate)	0.10	0.024	0.67	0.19
Thermotropic liquid crystal polymer	0.086	0.026	0.63	0.19
Poly(phenylene sulfide)	0.067	0.021	0.73	0.24
<i>Glass</i>				
Borosilicate glass (glass)	0	0	0	0

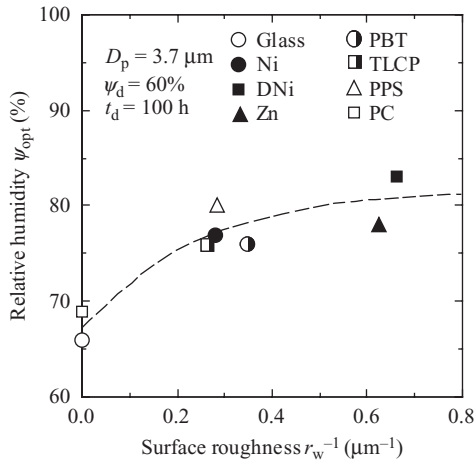


FIGURE 15.16 Optimum humidity for achieving maximum removal efficiency for various materials with different surface roughness.

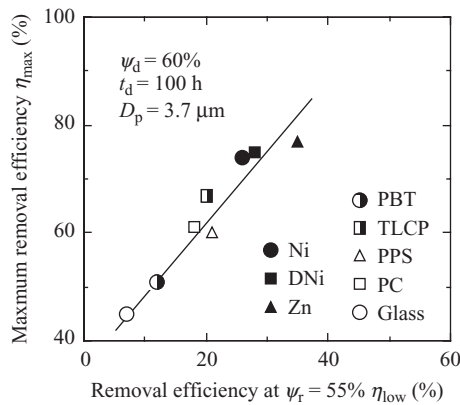


FIGURE 15.17 Correlation between maximum removal efficiency and the removal efficiency at low humidity.

properties and particle size on removal, and carried out theoretical analysis in which the surface properties are taken into consideration.

15.3 NEW REMOVAL METHODS USING THE AIR JET METHOD

15.3.1 Pre-Charging Method¹³

In the pre-charging method, the surface on which particles are adhered is charged unipolarly by the apparatus shown in [Figure 15.18](#) (i.e., as a pretreatment for the air jet removal, the surface is charged by a needle-type electrode

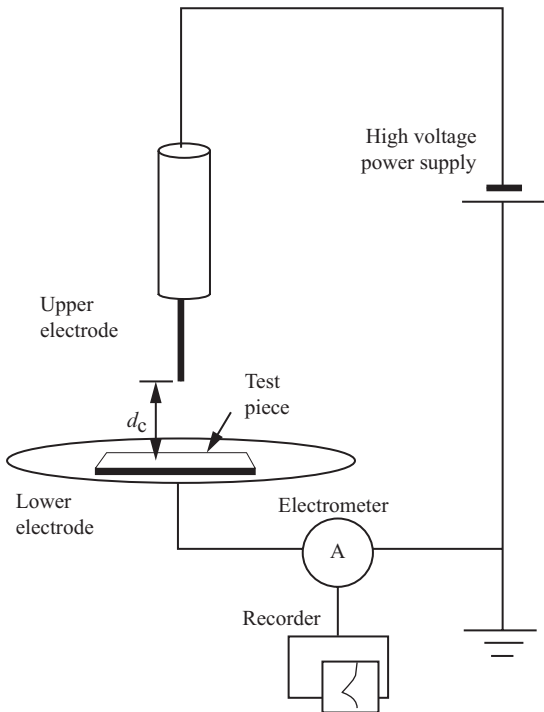


FIGURE 15.18 Apparatus for pre-charging the surface.

before applying a high-speed air jet). The upper electrode is set normal to the surface and the distance d_c between the tip of the electrode and the surface was maintained at 5 mm. The duration for applying the voltage was set at 10 min.

The experimental results for various applied voltages are shown in Figure 15.19. In this figure, the voltages denoted as V_c and $V_{\max}$ indicate the minimum voltage causing corona discharge and spark discharge, respectively. For the glass plate, the removal efficiency shows a maximum value of 43% at $V=5$ kV. This is four times higher than the efficiency obtained without pre-charging ($V=0$). The voltage for the maximum efficiency is lower than the critical voltage V_c . This suggests that corona discharge does not occur at the optimum voltage for the glass plate and only the non-uniform electric field is formed.

On the other hand, for the copper surface, the removal efficiency increases with increasing applied voltage. Spark discharge occurs when the voltage is higher than 15 kV and stable charging is impossible for the copper surface. Therefore, the optimum voltage for copper is the maximum voltage $V_{\max}$ ($=15$ kV). In this case, the efficiency is about seven times higher than that obtained without pre-charging.

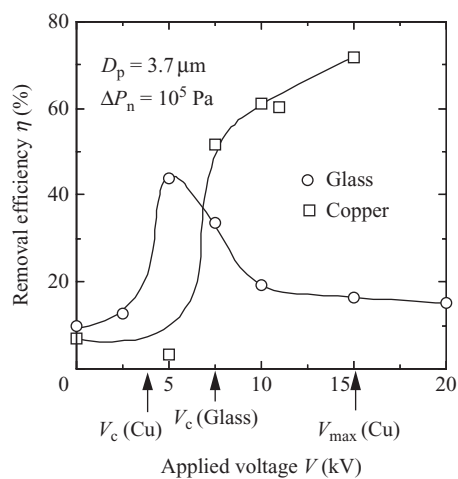


FIGURE 15.19 Removal efficiencies for glass and copper surfaces obtained by the pre-charging method.

After the pre-charging, agglomerated particles were found on the surface, as shown in Figure 15.20. We have confirmed that all the particles deposited are fully dispersed on the surface before pre-charging. This means that the agglomerates were formed by the pre-charging. The formation of the agglomerates is dependent on the type of surface. Spherical agglomerates are formed on the glass surface, while accumulated layers are formed on the copper surface.

After pre-charging, the number of agglomerates is counted and the ratio of the number of agglomerates to the total number of primary particles deposited (defined as the agglomeration ratio R_a) is obtained. Figure 15.21 shows the correlation between the agglomeration ratio R_a and the increment of removal efficiency $\Delta\eta$ ($=\eta - \eta(V=0)$). For the glass surface, the increment $\Delta\eta$ is proportional to the agglomeration ratio R_a . In other words, the ratio of agglomeration caused by pre-charging is equal to the increment of the efficiency by pre-charging. Therefore, it can be concluded that the increment for the glass plate was achieved by removing the agglomerated particles formed by pre-charging. On the other hand, for the copper surface, the increment $\Delta\eta$ is always greater than the agglomeration ratio R_a , suggesting that pre-charging enhances the removal of not only agglomerates but also primary non-agglomerated particles. This may be attributable to the impaction of the removed agglomerated particles against other surface particles and/or decrease in the adhesion force between particle and surface.

Furthermore, we have studied the effect of the number density of particles on the surface and it was confirmed that the pre-charging effect on the removal efficiency increases with the number density. Clearly, the agglomeration of particles increases with the density and the pre-charging effect increases.

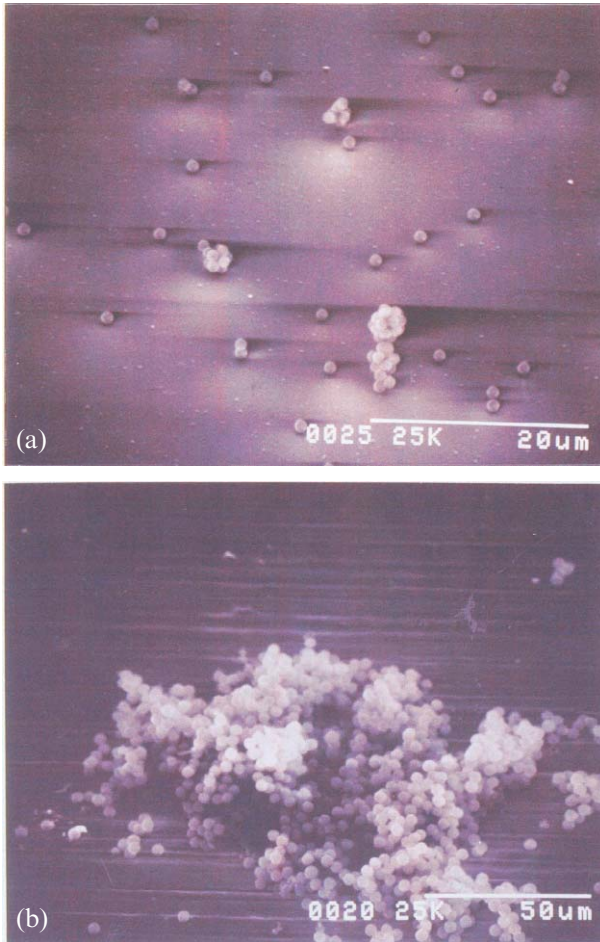


FIGURE 15.20 Photographs showing the deposited particles after pre-charging for (a) glass ($V=5$ kV) and (b) copper ($V=12$ kV) surfaces.

15.3.2 Vibrating Air Jet Method¹³

Figure 15.22 shows the newly designed vibrating-type air jet nozzle. It has a vibrating metal plate at the tip. Compressed air introduced into the nozzle causes vibration of the metal plate and generates a vibrating air jet. The width of the nozzle tip is 10 mm and the effective vibrating length l_v of the plate is 10 mm. The plate is fixed with a bolt, and the plate can be replaced with a different plate. In this experiment, carbon tool steel SK-3 (JIS G 4401) or quenched SK-3 was used as the vibrating plate. The thickness of the plate is 0.2 mm for both materials.

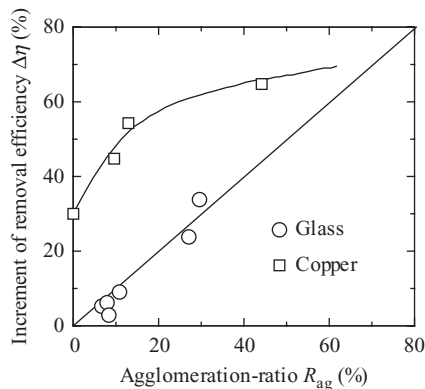


FIGURE 15.21 Correlation between the particle agglomeration ratio and the increment of the removal efficiency for glass and copper surfaces.

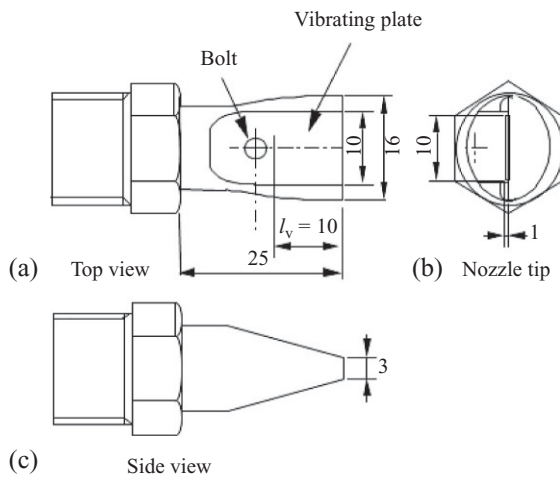


FIGURE 15.22 Schematic diagrams of the vibrating air jet nozzle showing (a) the top view, (b) the nozzle and the tip, and (c) the side view (dimensions are in millimeters).

The vibration of the air jet has an audible frequency. As shown in [Figure 15.23](#), the vibration generated by the nozzle consists of a fundamental frequency (i.e., the lowest frequency and the harmonics). Here, we have defined the frequency having the highest intensity as the representative frequency f of the vibration. The frequency f changes with the compressed air pressure ΔP_n . When the pressure ΔP_n is below 2.0×10^5 Pa, no clear vibration but a Gaussian noise was obtained. When ΔP_n is above 4.0×10^5 Pa, the air jet has no vibration. Therefore, for the vibrating nozzle, the applied nozzle pressure was ~ 2.0 – 4.0×10^5 Pa. The frequency values for the minimum and maximum nozzle pressure are listed in [Table 15.2](#).

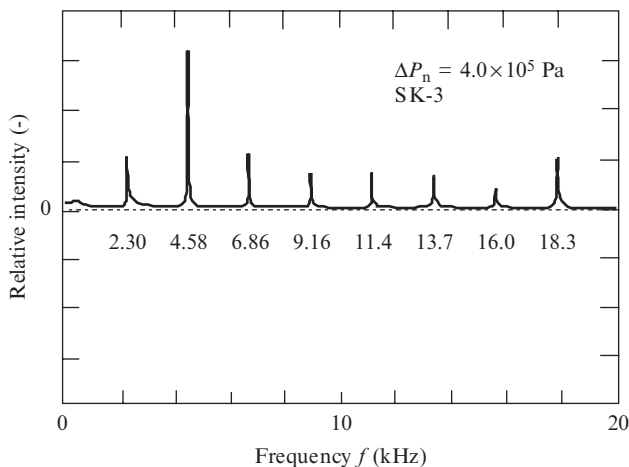


FIGURE 15.23 Power spectrum of the vibrations generated by the vibrating air jet nozzle.

TABLE 15.2 Range of Vibration Frequencies for the Vibrating Air Jet Nozzle		
ΔP_n (Pa)	2.3×10^5	4.0×10^5
SK-3 (kHz)	2.48	4.58
SK-3 with quenching (kHz)	2.53	4.78

Figure 15.24 shows the experimental results of the vibration air jet removal for particles with $D_p = 1.09 \mu\text{m}$ at various nozzle pressures. Although it was almost impossible for the standard nozzle to remove these particles under any nozzle pressure tested with $d = 10 \text{ mm}$, a removal efficiency of 76% was achieved by the vibrating air jet under the same operating conditions. The result shows that the vibration air jet is more effective than the standard air jet.

Pre-charging and jet vibration in series were also tested (i.e., after drying a particle-deposited surface, pre-charging is applied and then removal is carried out by using the vibrating air jet). A borosilicate glass plate was used as the test surface and the applied voltage was set at $V = 5 \text{ kV}$ to attain the maximum efficiency in Figure 15.19. Here, the vibrating plate SK-3 was used and the nozzle pressure was set at the maximum pressure $\Delta P_n = 4 \times 10^5$. The removal efficiency was 87.1%, which is almost 10% higher than that obtained by the vibrating air jet without pre-charging.

We have also tried vibrating air jet removal while simultaneously applying corona charging. The removal efficiency obtained by this method is only 4% higher than the vibrating air jet without pre-charging. This suggests that the

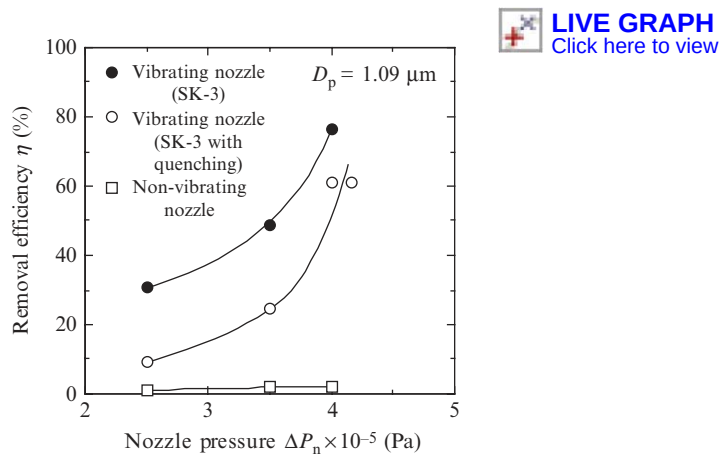


FIGURE 15.24 Removal efficiency achieved with the vibrating air jet as a function of the nozzle pressure.

airflow interferes with agglomeration caused by the pre-charging and the pre-charging is less effective in this system.

As described above, the vibrating air jet is effective for particle removal. However, the effect of frequency f and nozzle pressure ΔP_n on the efficiency is very complex.¹³ The optimum vibration of the air jet has not yet been obtained.

15.3.3 Other Removal Methods

One of the other methods is removal by a pulsed air jet. The experimental conditions reported in the literature are summarized in Table 15.3. Masuda et al.³ reported that the pulsed air jet did not significantly enhance particle removal. In contrast, Otani et al.¹⁴ found that the removal efficiency increased with increase in the number of pulsed jets. Ziskind et al.⁹ also found that there is a marked enhancement of the removal efficiency at a certain frequency; below and above this frequency the removal efficiency is lower. The differences in the results may be attributed to the difference in the nozzle shape and the geometric parameters of setting.

Figure 15.25 shows comparison of removal efficiencies obtained by a pulsed jet and a normal jet generated by two types of jet nozzles.¹⁵ In the case of Nozzle B and a particle with $D_p = 4.7 \mu\text{m}$, there is no difference in the removal efficiencies obtained by the pulsed jet and the normal jet. However, when they were applied to a particle with $D_p = 3.25 \mu\text{m}$ the removal efficiency obtained by the pulsed jet is higher than that obtained by the normal jet. In case of Nozzle D and a particle with $D_p = 2.25 \mu\text{m}$, the pulsed jet enhanced the removal efficiency.

TABLE 15.3 Experimental Conditions for Pulsed Air Jet Removal from Published Studies

	Author(s)		
	Masuda et al. ³	Otani et al. ¹⁴	Ziskind et al. ⁹
Nozzle exit shape	Rectangle	Rectangle	Circle
Surface material	Glass	Glass, silicon	Glass, silicon
Particle material	Styrene/ divinylbenzene	Polystyrene latex	Alumina silicate
Particle diameter (μm)	1.09–11.9	0.25–1.1	2.0–5.0
Distance from nozzle (mm)	3.0–25.0	6.0	30.0
Impinging angle ($^\circ$)	15–45	30	20–50
Duration of pulse (s)	1.0	1.0	0.0078– 0.0047
Pulse interval (s)	1.0	3.0	0.0078– 0.0047
Frequency (Hz)	0.5	0.25	64–107

Here, we defined the index I of removal efficiency improvement by pulsed jet as the difference in the nozzle pressures giving the same removal efficiencies ($=\Delta P_{\text{normal}} - \Delta P_{\text{pulsed}}$; ΔP_{normal} is the pressure drop ΔP_n for a normal air jet and ΔP_{pulsed} is the pressure drop ΔP_n for a pulsed air jet giving the same removal efficiency as removal by a normal air jet). The index I is shown in Figure 15.26 as a function of the nozzle pressure in normal jet ΔP_{normal} . The index I increased from a certain nozzle pressure. The pressure at which the index I starts to increase is independent of the nozzle shape and tested particle diameter. The pressure is almost the same as the critical pressure P_{nc} [see Equation (15.3)] in which the air jet speed at the nozzle tip reaches the sonic speed. In other words, the improvement in removal efficiency by pulsed air appears only when the nozzle pressure is greater than the critical pressure P_{nc} .

When the nozzle pressure is set above the critical pressure, the improvement in removal efficiency is maximum at a certain pulse frequency (=optimum frequency), as shown in Figure 15.27. Measurement of the dynamic pressure at the surface of a flat plate revealed that the air pressure never falls to zero when the pulse frequency is above the optimum frequency. Ziskind et al.⁹ suggested that the pulsed air jet brings the flow field near the surface to its initial state, and the velocity inside the boundary layer becomes high once again. Therefore, the

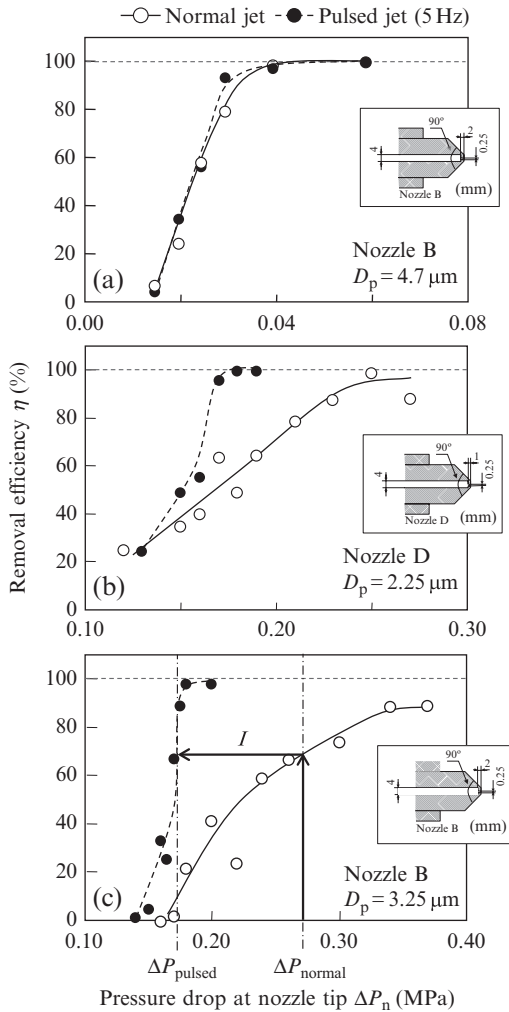


FIGURE 15.25 Comparison of removal efficiencies obtained by pulsed jet and normal jet for particle size, D_p , of (a) $4.7 \mu\text{m}$, (b) $2.25 \mu\text{m}$, and (c) $3.25 \mu\text{m}$.

force acting on the particles is restored to its maximum value for each pulse. This implies that the average force for the jet duration time increases with the pulse frequency and the impulse generated by each jet impingement increases with the pulse frequency. This explanation is based on the assumption that the pulses are independent. If the pulse is affected by the flow induced by the former pulse, the pulsed air jets may have almost the same effect as the continuous normal jet. Thus, an optimum frequency is achieved.

Another method studied by Smedley et al.¹⁶ is the removal of particles by impinging shock waves. The shock waves are generated in an open-ended shock

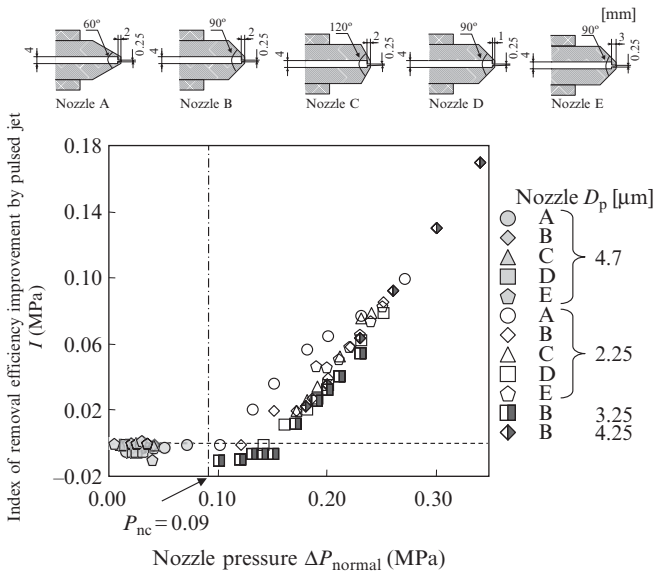


FIGURE 15.26 Index I of removal efficiency improvement by pulsed jet as a function of nozzle pressure for normal jet obtained from particle sizes, D_p , of 2.25, 3.25, 4.25, and 4.7 μm and five types of nozzles having different nozzle tip shapes.

 **LIVE GRAPH**
Click here to view

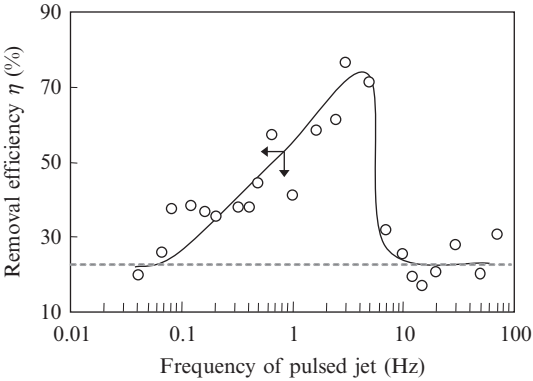


FIGURE 15.27 Removal efficiency obtained by pulsed air jet as a function of frequency of pulsed jet.

tube, thus producing ultrasonic velocity pulses with large time intervals. The authors focused on the removal length in the direction aligned with the long axis of the removed area where it has a circular (normal impingement) or an elliptic shape (inclined impingement). The length increased with shock number. This means that the number of pulsed jets enhances the removal.

15.4 REMAINING PROBLEMS

When we consider the industrial process required for the removal of particulate contaminants, the process can be divided into three stages, namely contamination, transfer from the contaminated zone to the removal apparatus, and removal of the contaminants. This means that there are three environments. Thus, we have investigated the effect of humidity on the deposition of particles and the time to transfer particles.¹⁰ Here, the humidity of the transferring environment was kept constant and very low (15%) using a desiccator. In addition, the transfer time was taken into account as the drying time t_d in the desiccator.

Figure 15.28 shows the results of the removal experiment by varying the drying time in the desiccator and taking the humidity ψ_d during the deposition of particles as a variable parameter. The humidity ψ_r at removal was kept constant at 59%. When the humidity at deposition was lowest ($\psi_d=55\%$), the removal efficiency was reduced as the drying time was increased, and reached a constant value of about 10% after 5 h of drying. At $\psi_d=58\%$, the reduction in the removal efficiency was minor, until about 20 h of drying when a high removal efficiency of 70–80% was obtained; the efficiency rapidly decreased after the drying time exceeded 20 h. When the drying time exceeded 60 h, the efficiency was equal to that at $\psi_d=55\%$ and reached a constant value. At ψ_d values of 64% and 72%, although the removal efficiency increased together with the drying time until about 3 and 6 h of drying respectively, it dropped rapidly thereafter. In other words, (a) the removal efficiency decreased as the drying time was increased for the case when the deposition took place at a humidity not higher than the humidity ψ_r at removal; (b) the removal efficiency increased at the beginning and decreased after the maximum efficiency was achieved at a certain drying time for the case when the deposition took place at a humidity not lower than that at removal; and (c) the removal efficiency was not affected by the humidity ψ_d at deposition after 80 h of drying. As a result, it is suggested that the removal efficiency is higher immediately after

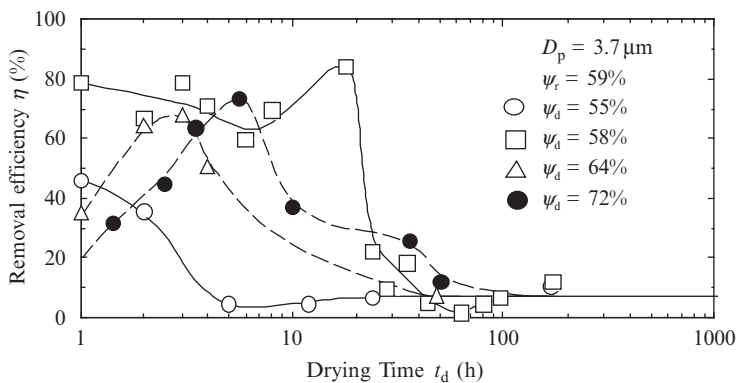


FIGURE 15.28 Effect of relative humidity and drying time on the removal efficiency.



the deposition of particles for the case when the particles are deposited at a relatively low humidity and, after drying them to a certain level, for the case when the particles are deposited at high humidity.

In order to avoid this effect, over 80 h of drying time was needed. This is the reason why the test piece was dried for 100 h. It was also confirmed by other experiments that the removal efficiency is not affected by the residence time in the environment if the time is longer than 1 min. Therefore, the test piece was transferred to a controlled removal environment and the removal experiment was conducted after leaving the test piece in the environment for 2 min or longer.

The mechanism of the effect of the deposition environment is completely unknown. The adsorption layer of water molecules must be one of the main causes of the effect. The effect cannot be explained by adsorption alone because it is recognized that the adsorption phenomena is a fast phenomenon of the order of 1 s or less. The present results certainly reflect the change in the adhesion force with time. The time dependence of the adhesion force has not yet been established. This problem is not limited to the removal of particles using a high-speed air jet.

15.5 SUMMARY

In this chapter, we have summarized the empirical knowledge of particle removal using a high-speed air jet, as the basic principle of the removal or the resuspension phenomena are not well understood. Although we have shown how the removal efficiency changes with operating conditions, the mechanisms of the effects of the operating conditions are unclear. The experiments were conducted with micrometer-sized particles. However, the removal results for sub-micrometer particles have already been reported.¹⁷ If the mechanisms of the removal were well established and the optimum operating conditions were obtained, it is expected that the impinging air jet removal method can be applied to precision removal of solid particles of sub-micrometer size.

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